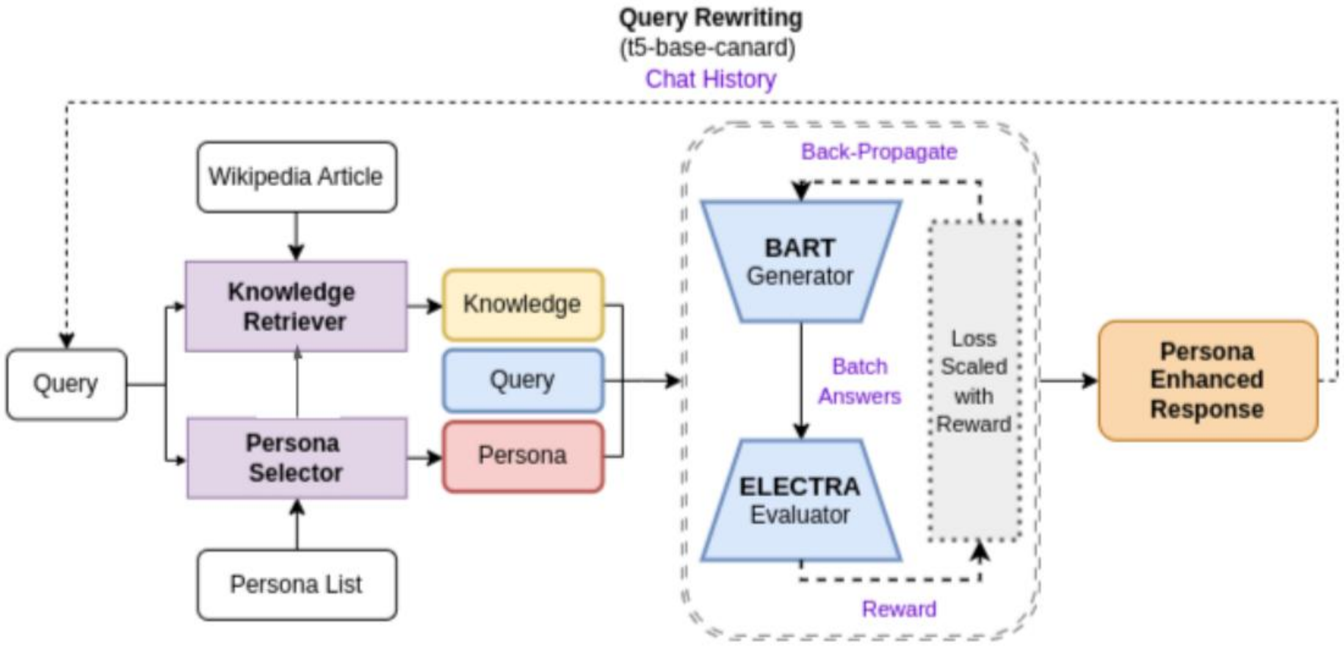
DH 604

Persona based Proactive Conversational Agent

04-02-2026

Personalized Response Generation using Dynamic Knowledge Retrieval and Persona Adaptive Queries

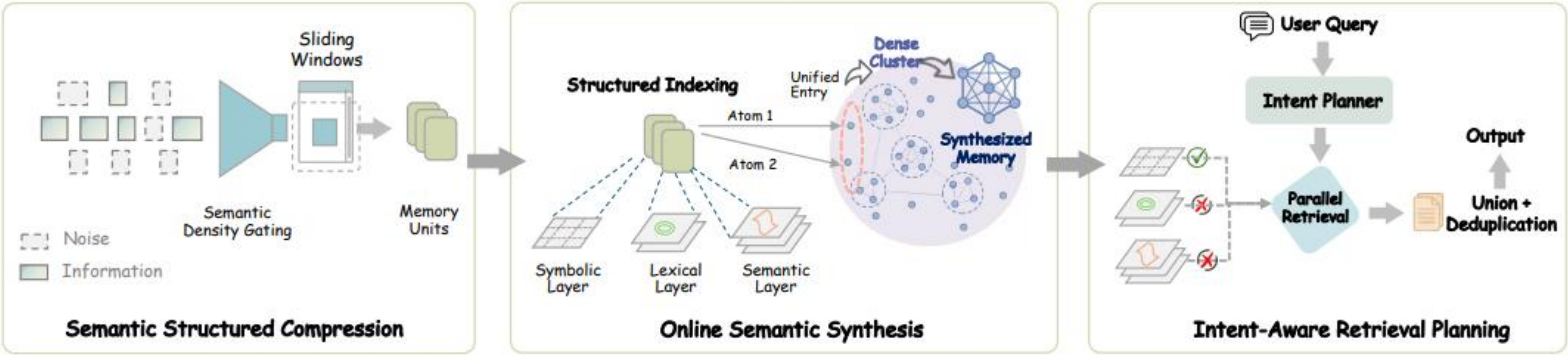


1. The model employs **Dynamic Knowledge Retrieval** to supplement responses with information from background sources, using Dense Passage Retrieval (similar to RAG).

2. A dedicated **Persona Selector** module chooses a persona that aligns with the user's query. This is modeled as a multi-label classification task, which includes an option for "no-persona" for generic queries.

1. Query Rewriting: Utilizes a LLM model to refine the user's input based on chat history.
2. Response Generation: A **BART** generator is paired with an **ELECTRA** evaluator.
3. Reward Function: The system uses a balancing reward function that considers generative capabilities like coherence to ground truth responses and persona alignment.

SimpleMem: Efficient Lifelong Memory for LLM Agents



1. **Semantic structured compression**: This stage active filters redundant interaction content, such as social chit-chat or repetitive logs, using an entropy-based filtering. Temporal dialogue data are split into overlapping windows. Raw dialogue windows are evaluated based on the introduction of new entities and semantic novelty.
2. **Online Semantic Synthesis**: It is a recursive memory consolidation step. Units are grouped into clusters based both semantic similarity and temporal proximity.
3. **Intent-aware retrieval**: This mechanism dynamically adjusts the retrieval scope based on the estimated complexity of the user query. A lightweight classifier determines if a query requires a simple fact lookup or complex multi-step reasoning. Retrieval depth (*k*) is modulated accordingly, ranging from a minimum of 3 entries for simple queries to a maximum of 20 for complex ones, ensuring a balance between accuracy and token efficiency. This is done using a reasoning LLM

Target-Guided Conversations

1. The goal of target-guided dialogues is to achieve the designated target effectively and efficiently through engaging conversations

2. **Topic-shift Detection:** Discover the topic drift in the user utterances during the conversation. Fine-tune sentence classifier to classify the utterances into major, minor and off topics for detecting the digression from the target topic of the conversation.

3. **Topic Planning:** Enables the conversation to follow an expected topic direction. Can leverage external knowledge graphs for improving the quality of topic transitions with graph reasoning techniques. -> *GraphRAG retrieval technique*

4. **Topic-guided Response Generation:** Based on the topic/keyword selected by the planner, augment the user query to generate topic related response

Profile Building

1. Objective is to build a persona profile from user dynamically as conversation/ journalling continues

2. **Medical Persona Ontology :** Specify keys/ set of features we want to track. Eg: sleep quality, diet pattern, exercise frequency

3. **Persona-Extractor:** Can be LLM based methods to extract health-related facts into predefined schema or rule based like Named Entity Recognition

4. **Continuous Updates:** As conversation moves, keep on updating the profile. Store statistics to capture trends, this can be later used for task planning

5. **Use Persona for Consistency:** This profile can also be used to create topic planning.

① RAG

② How to evaluate the RAG

③ Papers

④ Dataset

⑤ Framework

$\boxed{\text{Raw data}}$ → Personas/Attributes

↓

Find relations

problems → link to other attributes

Rule based → | e_1

$\alpha \cdot e_1 + (1-\alpha)\, e_2$

↓

$\boxed{\text{Reasoing}}$ →

↓

$\boxed{\text{Task scheduler}}$

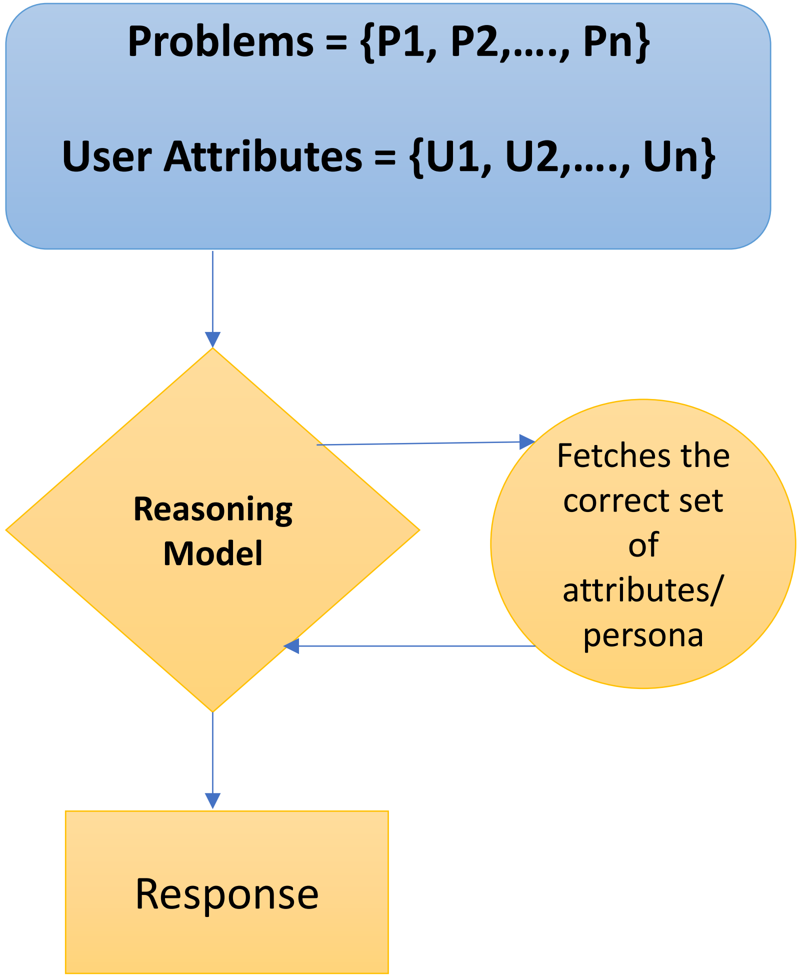
DH 604

Persona based Proactive Conversational Agent

11-02-2026

Retrieval & Persona Grounding

- **Objective is to generate responses aligned to specific persona**
- **Need an evaluation metric for judging the RAG responses**
- **This will evaluate whether the response is:**
  - **Correct with respect to the knowledge provided**
  - **Is aligned with the correct set of persona**
  - **Also solves the problem of persona selection**



FoCus Dataset

This dataset contains:

1. Multi turn dialogues

2. User persona/ attributes

3. Persona grounding for each response

   - For persona {P1, P2, .. , Pn}
   - Grounding: {0, 0, 1,…,}

We use this data to train a classifier which will take input as:

1. Conversation History
2. Question
3. Answer
4. Persona

And will return if the answer for the question is grounded to the particular persona

This will help us evaluate if out retrieval is able to select persona.

We can finetune a sentence classifier for this purpose

If the classifier is C, then score (probability) is ***Score1 = C(H, Q, A, P)***
Response correctness can be evaluated by BERT Score. So ***Score2 = BERT Score***

Score = x. Score1 + (1-x). Score2